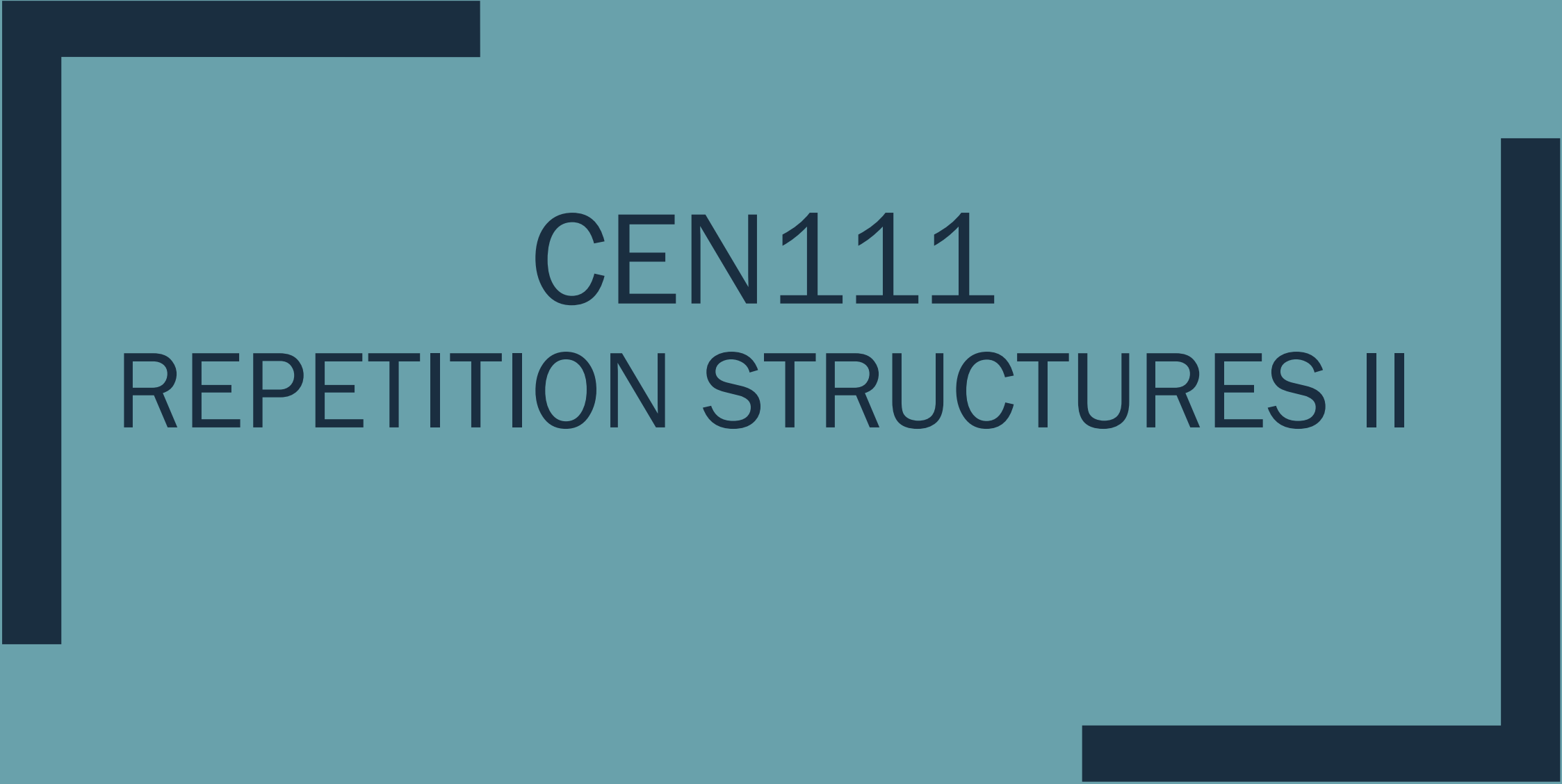


The slide features a light blue background with two dark blue L-shaped brackets. One bracket is in the top-left corner, and the other is in the bottom-right corner. Centered between them is the text 'CEN111' and 'REPETITION STRUCTURES II' in a dark blue, sans-serif font.

CEN111

REPETITION STRUCTURES II

Structures

- Sequence structures:
 - *Programs executed sequentially by default*
- Selection structures:
 - *if, if...else, and switch*
- Repetition structures:
 - *while, do...while and for*

do...while Iteration Statement

```
do
{
    statement;
}
while (loop repetition condition);
```

do...while Iteration Statement

- The do...while iteration statement is similar to the while statement.
- In the while statement, the loop-continuation condition is tested at the beginning of the loop before the body of the loop is performed.
- The do...while statement tests the loop-continuation condition after the loop body is performed.
- Therefore, the loop body will be executed at least once.
- When a do...while terminates, execution continues with the statement after the while clause.

do...while Iteration Statement

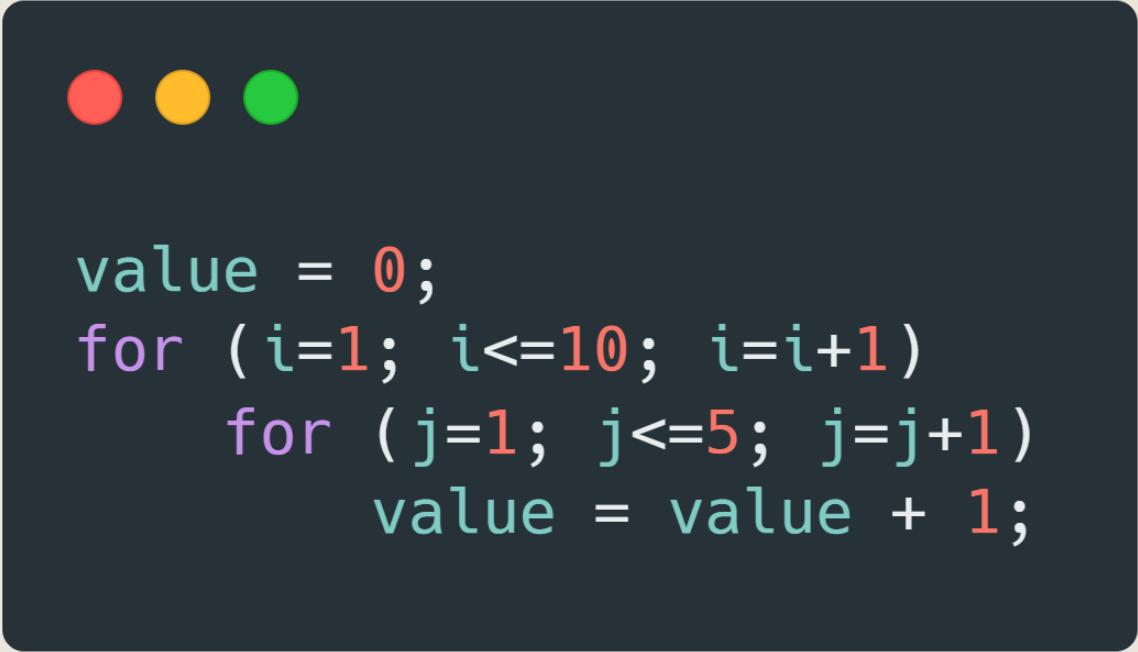


```
#include <stdio.h>
int main()
{
    int counter = 1;
    do
    {
        printf( "%d ", counter );
        counter = counter + 1;
    } while ( counter <= 10 );
    return 0;
}
```

Nested Loops

- When a loop body includes another loop construct this is called a nested loop.
- In a nested loop structure the inner loop is executed from the beginning every time the body of the outer loop is executed.

Nested Loops



```
value = 0;  
for (i=1; i<=10; i=i+1)  
    for (j=1; j<=5; j=j+1)  
        value = value + 1;
```

- How many times the inner loop is executed?

Nested Loops



```
#include <stdio.h>

int main()
{
    for (int i=1; i<=5; i++)
    {
        for (int j=1; j<=i; j++)
            printf("*");
        printf("\n");
    }
    return 0;
}
```

The `break` and `continue` Statements

■ `break`

- *Causes immediate exit from a while, for, do...while or switch statement*
- *Program execution continues with the first statement after the structure*

■ Common uses of the `break` statement

- *Escape early from a loop*
- *Skip the remainder of a switch statement*

The break and continue Statements



```
#include <stdio.h>
int main(void)
{
    int x;
    for(x = 1; x <= 10 ; x++)
    {
        if( x == 5)
            break;
        printf("%d ", x);
    }
    printf("\nBroke out of the loop at x=%d", x);
    return 0;
}
```

The `break` and `continue` Statements

- `continue`

- *Skips the remaining statements in the body of a while, for or do...while statement*
- *Proceeds with the next iteration of the loop*

- `while` and `do...while`

- *Loop-continuation test is evaluated immediately after the `continue` statement is executed*

- `for`

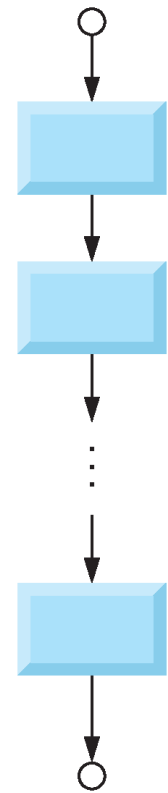
- *Increment expression is executed, then the loop-continuation test is evaluated*

The break and continue Statements

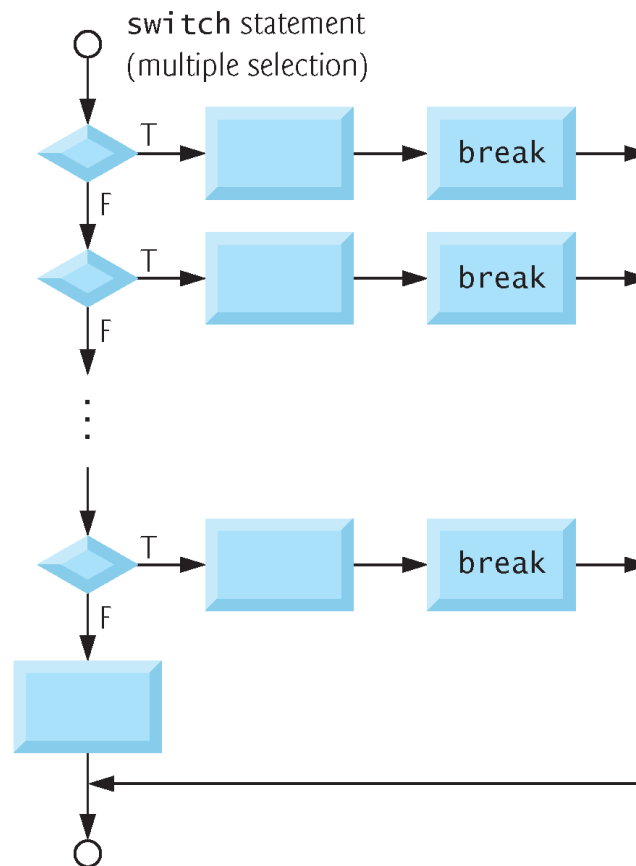
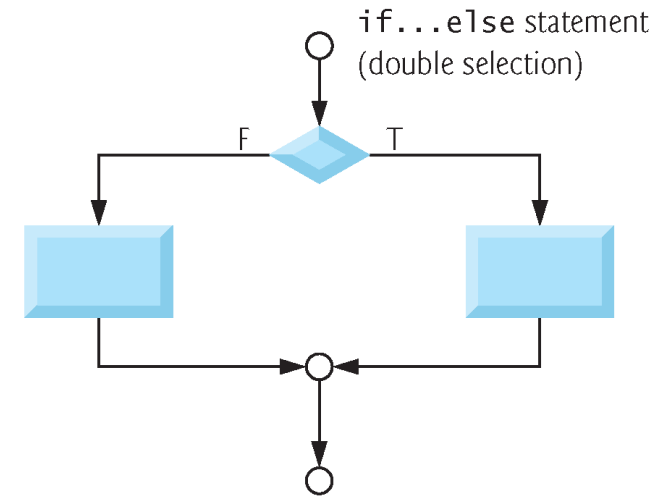
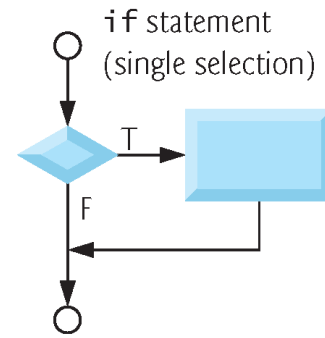


```
#include <stdio.h>
int main(void)
{
    int x;
    for(x = 1; x <= 10 ; x++)
    {
        if( x == 5)
            continue;
        printf("%d ", x);
    }
    return 0;
}
```

Sequence

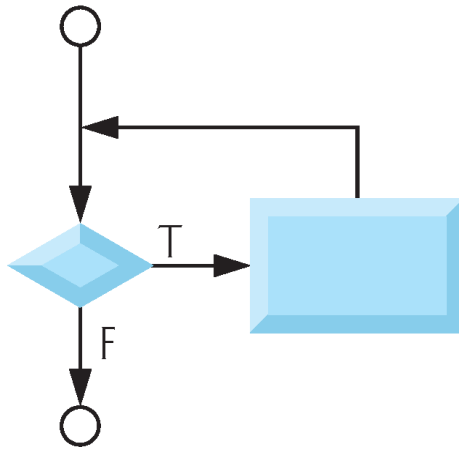


Selection

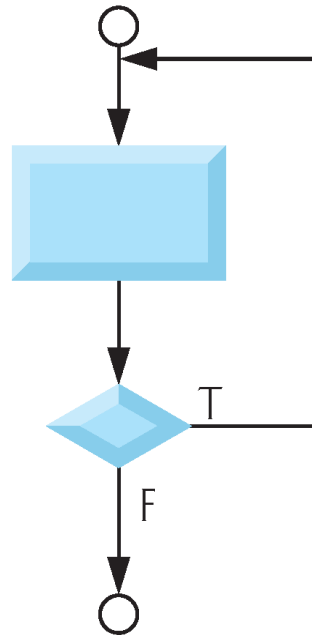


Repetition

while statement



do...while statement



for statement

